SULIT

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER I 2023 / 2024

SUBJECT CODE	**: SECR2043 OPERATING SYSTEMS**
SECTION	**: 1, 2, 5, MJIIT**
TIME	**: 9.00 AM – 12.00 PM**
DATE/DAY	**: 16 FEBRUARY 2024**
VENUES	**: BK 1-6, N24**

INSTRUCTIONS:

Answer all questions from PART A, and PART B in the answer booklet. Show all your works.

This test will contribute **35%** towards the total marks of 100%.

(Please write your particulars and circle you section)

Name	
I/C No.	
Year / Course	
Section / Lecturer Name (Circle)	**[01] - Dr Farkhana Binti Muchtar** **[02] - Dr Farkhana Binti Muchtar** **[05] - Dr Farkhana Binti Muchtar** **[MJIIT] – Dr. Kaiyisah Hanis Binti Mohd Azmi**

This question booklet consists of 18 pages including the front page.

PART A: OBJECTIVES

Answer ALL questions in the answer booklet.

1. Which of the following is NOT a necessary condition for deadlock?
   A. Mutual Exclusion.
   B. Hold and Wait.
   C. Circular Wait.
   D. Resource Sharing.

2. Which of the following is a characteristic of the Hold and Wait condition?
   A. A process must be holding at least one resource and waiting to acquire additional resources.
   B. Resources are divided into several types.
   C. Resources must be used in a mutually exclusive manner.
   D. Processes must release resources after a fixed period.

3. A resource allocation graph is used in deadlock handling to:
   i. Show the current allocation of resources.
   ii. Indicate processes waiting for resources.
   iii. Demonstrate the presence of a cycle.
   iv. Calculate the maximum resources needed by each process.
   A. i, ii, and iii
   B. i and iii
   C. ii, iii, and iv
   D. i, ii, iii, and iv

4. In fixed partitioning, if all partitions are occupied and a new process arrives, the OS must:
   A. Overwrite an existing partition.
   B. Increase the partition size.
   C. Wait for a partition to become free.
   D. Store the process in virtual memory.

5. Deadlock Avoidance differs from Deadlock Prevention in that:

   i. Avoidance requires knowledge of future requests.
   ii. Prevention eliminates one of the necessary conditions for deadlock.
   iii. Avoidance uses algorithmic strategies to ensure safe resource allocation.
   iv. Prevention typically leads to underutilization of resources.

   A. i and ii
   B. ii and iii
   C. i, iii, and iv
   D. All of the above

6.

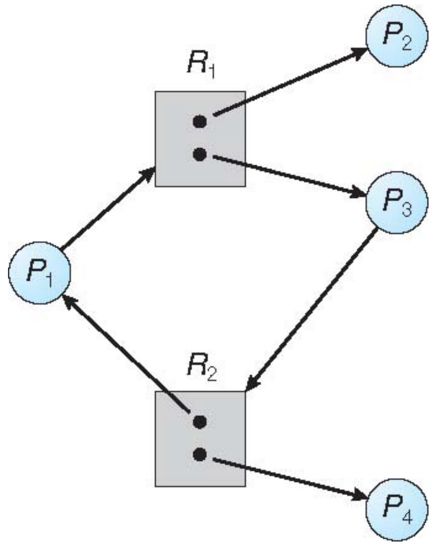

Figure 1

Evaluate the given resource allocation graph in **Figure 1** and choose the correct statement:

   i. Process P1 is holding one instance of resource R2 and is requesting one instance of resource R1.
   ii. Process P3 is waiting for a resource that is currently being held by P2.
   iii. The graph indicates a deadlock, as there is a cycle involving all the processes and resources shown.
   iv. Resource R1 is being held by two different processes, P2 and P3.

   A. i and ii
   B. ii and iii
   C. i and iv
   D. None of the above

7. Consider a system with three processes and three resources. The system is currently in the following state:

- Allocation: P0 (0, 1, 0), P1 (2, 0, 0), P2 (3, 0, 3)
- Maximum: P0 (7, 5, 3), P1 (3, 2, 2), P2 (9, 0, 2)
- Available: (3, 3, 2)

If P1 requests (1, 0, 2), should the request be granted?

A. Yes

B. No

C. Only if P2 releases some resources

D. The request is invalid since P1's total claim would exceed its maximum

8. In a paged memory allocation scheme, what does a page table store?

A. The actual data of the pages.

B. The fragmentation information of each page.

C. The frame number associated with each page.

D. The size of each page.

9. _________ occurs when allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used.

A. External fragmentation

B. Internal fragmentation

C. Segmentation fault

D. Paging error

10. The _________ algorithm scans memory from the beginning and chooses the first available block that is large enough to accommodate the process.

A. Best-fit

B. Worst-fit

C. First-fit

D. Next-fit

11. Consider **Figure 2**, which illustrates the dynamic address translation process employed in memory management.

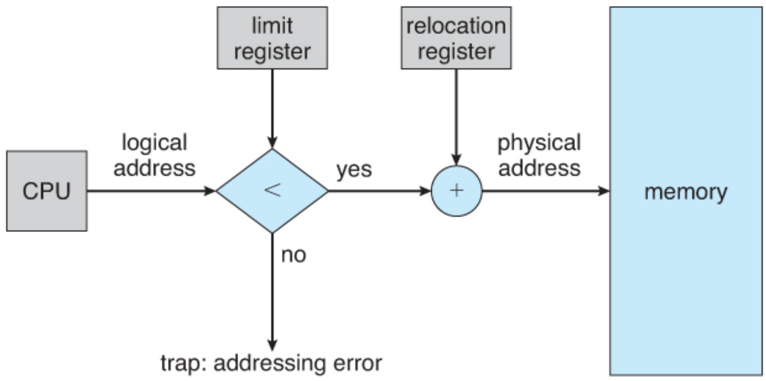

Figure 2

Select the statements that are true concerning this process:

i. The limit register is employed to verify that the logical address is within the process's allocated memory space.

ii. The relocation register contains the base address to which the logical address is added to derive the physical address.

iii. An addressing error trap is initiated if the logical address exceeds the limit set by the limit register.

iv. The memory management unit directly accesses the physical address after the logical address passes the limit check.

A. i and ii

B. i, ii, and iii

C. ii and iv

D. All of the above

12. What does 'dirty bit' in page table entries indicate?

A. Page has been replaced recently.

B. Page needs to be written to disk.

C. Page contains a syntax error.

D. Page is corrupted.

13. In the context of virtual memory, thrashing can be best described as:
    A. The system spends most of its time servicing page faults.
    B. Memory is frequently swapped between RAM and disk.
    C. A process is in a state of deadlock.
    D. The page replacement algorithm is inefficient.

14. Select the characteristics of swapping in memory management:

    i. It uses disk space as a temporary storage.
    ii. It requires contiguous memory allocation.
    iii. It is a solution to memory overcommitment.
    iv. It is faster than paging.

    A. i and iii
    B. ii and iv
    C. i and ii
    D. iii and iv

15. In a virtual memory system, ________ is a small hardware device that stores a limited number of page table entries for quick lookup.
    A. Page Table
    B. Translation Lookaside Buffer (TLB)
    C. Disk Controller
    D. Memory Register

16. Given four page frames and the reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, with the LRU page replacement algorithm, the number of page faults is:
    A. 6
    B. 7
    C. 8
    D. 9

17. In a file system, a 'file control block' (FCB) typically does NOT contain which of the following information?
    A. File permissions.
    B. File size.
    C. Memory address of the file.
    D. File owner.

18. Consider the following factors in a page replacement algorithm's effectiveness:
    i. Number of page frames available.
    ii. Frequency of page access.
    iii. Process execution patterns.
    iv. Size of the physical memory.
    A. i, ii, and iii
    B. ii, iii, and iv
    C. i, iii, and iv
    D. All of the above

19. The principle of locality in a virtual memory system implies that:
    i. Processes access a small subset of their pages frequently.
    ii. Most memory accesses occur within a limited temporal window.
    iii. Sequential memory access is more common than random access.
    iv. Working sets of processes change infrequently.

    A. i and ii
    B. i, ii, and iii
    C. ii, iii, and iv
    D. All of the above

20. Which of the following is a disadvantage of the linked allocation method in file systems?
    A. It requires more space for pointers in each block.
    B. It does not support direct access to file blocks.
    C. It causes external fragmentation.
    D. It has a higher risk of file corruption.

21. In a file system, ________ is a process that involves verifying and repairing inconsistencies in file system data structures.

A. defragmentation

B. formatting

C. consistency checking

D. compression

22.

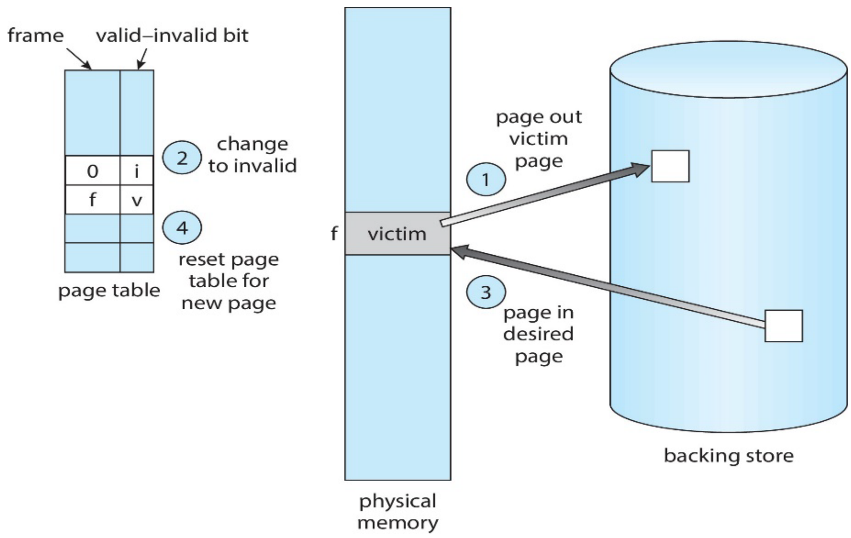

Figure 3

According to **Figure 3**, which of the following statements correctly describe the actions involved in loading a page into memory during page replacement?

i. The valid-invalid bit is set to invalid to indicate that the page is no longer in the physical memory.

ii. The victim page is selected and written back to the backing store if it has been modified.

iii. The desired page is retrieved from the backing store and loaded into the frame that held the victim page.

iv. After the page replacement, the page table is updated to reflect the new frame location of the desired page.

A. i, ii, and iii

B. ii, iii, and iv

C. i, iii, and iv

D. All of the above

23. In disk scheduling, the SSTF algorithm minimizes the:

    A. Total number of tracks moved.
    B. Time for a full rotation of the disk.
    C. Average seek time.
    D. Total number of requests.

24. Which of the following statements accurately assess the advantages or trade-offs between different allocation methods for data storage?

    i. Contiguous allocation supports direct access but can lead to external fragmentation.
    ii. Linked allocation is suitable for sequential access but inefficient for direct access.
    iii. Indexed allocation overcomes the limitations of both contiguous and linked allocations.
    iv. Multilevel indexing reduces the overhead of large index tables.

    A. i, ii, and iv
    B. i, iii, and iv
    C. ii, iii, and iv
    D. All of the above

25. The ______ algorithm in disk scheduling is known for servicing requests that are nearest to the current head position, potentially leading to starvation of some requests.

    A. FCFS
    B. SSTF
    C. SCAN
    D. C-LOOK

26. RAID _______ is known for its mirrored set without striping, providing high data redundancy and fault tolerance.

    A. 0
    B. 1
    C. 5
    D. 10

27. Which of the following RAID levels offers the best performance for write-intensive applications?

A. RAID 0

B. RAID 1

C. RAID 5

D. RAID 6

28. Refer to **Figure 4**, which illustrates the indexed allocation method used in file systems. The index block number 19 for the file "jeep" contains several entries.

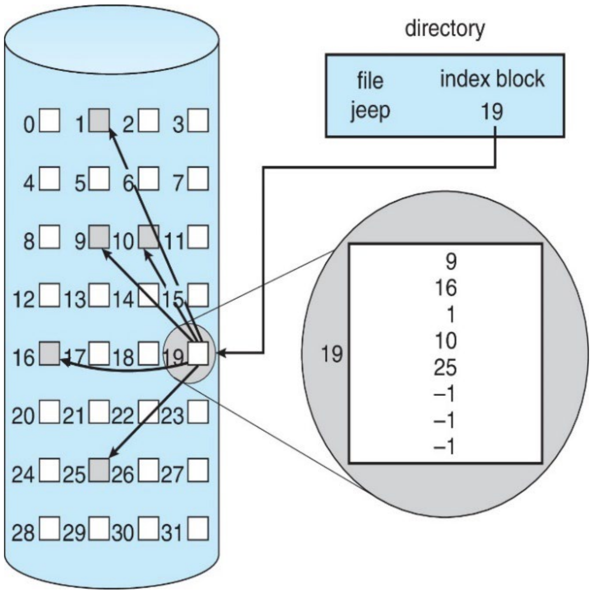

Figure 4

Which of the following statements is true regarding the structure and content of this index block?

A. Each entry in the index block corresponds to a direct pointer to a data block on the disk.

B. The last two entries of the index block indicate the file "jeep" occupies more blocks than listed.

C. The index block provides a sequential list of all the data blocks used by the file "jeep."

D. The entry with the number "25" suggests that the file "jeep" is fragmented across the disk.

29. Assess the following statements about disk scheduling algorithms:

   i. LOOK algorithm minimizes the total head movement.
   ii. SSTF can lead to indefinite postponement or starvation.
   iii. FCFS is the most efficient in terms of overall disk performance.
   iv. SCAN algorithm treats the disk as a circular list.

   A. i and ii
   B. i, ii, and iv
   C. ii, iii, and iv
   D. i, iii, and iv

30. Which of the following statements accurately describes the differences between magnetic disks and solid-state drives (SSDs) in terms of their storage characteristics?

   i. Magnetic disks have lower access times than solid-state drives (SSDs).
   ii. SSDs do not have moving parts, reducing their seek time.
   iii. Magnetic disks are generally more expensive per unit of storage than SSDs.
   iv. SSDs use electrical circuits to store data, leading to faster data transfer rates.

   A. ii and iv
   B. i, ii, and iii
   C. ii, iii, and iv
   D. i, iii, and iv

PART B: STRUCTURED

Answer ALL questions in the answer booklet.

QUESTION 1 [18 MARKS]

(a) Consider a system with dynamic partition memory management at time t0 as illustrated in **Figure 5**.

At time t1, Job B and Job C complete execution and free their memory. Subsequently, new jobs arrive: Job E (30 KB) and Job F (15 KB) at t2, followed by Job G (10 KB) at t4.

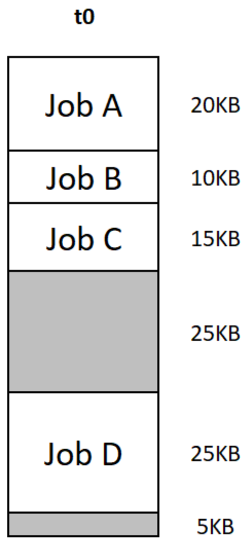

Figure 5

Draw the diagrams to visualize the memory configuration at t1, t2, t3 and t4. Apply merging or compaction strategies to address external fragmentation. [6 marks]

(b) Given six memory partitions as illustrated in **Table 1** below:

Table 1

Partition	**1**	**2**	**3**	**4**	**5**	**6**
Size (KB)	350	650	300	250	550	150

Consider five processes (P1, P2, P3, P4, P5) with different sizes needing to place in the memory (in order) as shown in Table 2(a) using the first-fit, best-fit, and worst-fit algorithms. Reconstruct and complete Table 2(b) by writing the memory partition numbers for each algorithm, respectively. [4 marks]

Table 2(a)

Processes	Size (KB)
P1	125
P2	475
P3	255
P4	150
P5	350

Table 2(b)

Process	First-Fit	Best-Fit	Worst-Fit
P1			
P2			
P3			
P4			
P5			

(c) Consider a paging system for a logical address space of 2^5 bytes with a page size of 2^3 bytes and a physical memory of 128 bytes. Answer the following questions. Show your working.

i. What is the frame size of the physical memory? [2 marks]

ii. How many frames of the physical memory? [2 marks]

iii. How many number of entries in the page table? [2 marks]

iv. Assume the frame number for a data is 8 with offset 1. Calculate the physical address for the data. [2 marks]

QUESTION 2 [19 MARKS]

Evaluate a system's state using the **Banker's Algorithm**. The system has **4 processes (P0 to P3)** and **3 resource types (A, B, C)**.

- **Total Resources:** A (12), B (8), C (6)
- **Available Resources:** A=5, B=3, C=2
- **Processes' Allocation and Maximum Requirements:**

Table 3

Processes	**Allocation**			**Maximum**		
	A	**B**	**C**	**A**	**B**	**C**
P0	1	2	1	3	4	2
P1	0	1	2	1	2	3
P2	3	0	1	4	1	2
P3	1	2	0	2	4	3

Provide detailed answer:

a) Compute the 'Need' matrix. [6 marks]

b) Determine if the system is in a safe state. [5 marks]

c) If safe, find the safe sequence. [2 marks]

d) Assume P1 requests an additional (0, 2, 0). Check if the system can allocate these resources. [3 marks]

e) For P3's request (1, 0, 2). heck if the system can allocate these resources [3 marks]

QUESTION 3 [16 MARKS]

A disk has 200 cylinders, numbered from 0 to 199. The disk queue has the following cylinder requests:

Cylinder Request Queue: 45, 170, 75, 130, 10, 55, 90, 150, 60

The current head position is at cylinder **100**, moving towards higher-numbered cylinders.

(a) Using the **FCFS (First-Come, First-Served) Disk Scheduling Algorithm**, draw disk scheduling graph and calculate the total movement of the disk arm. Show the point-to-point calculations. [5 marks]

(b) Using the **SSTF (Shortest Seek Time First) Disk Scheduling Algorithm**, draw disk scheduling graph and calculate the total movement of the disk arm. Show the point-to-point calculations. [5 marks]

(c) Using the **SCAN Disk Scheduling Algorithm (Outer->Inner)**, draw disk scheduling graph and calculate the total movement of the disk arm. Show the point-to-point calculations. [5 marks]

(d) In the given disk scheduling scenario, which algorithm demonstrates the most efficient performance? [1 marks]

QUESTION 4 [17 MARKS]

Given the following reference string:

Reference String: 3 1 4 2 5 3 4 1 6 2 5 4 3 1

(a) Calculate the number of page faults that will occur for each of the following page replacement algorithms with **3-page frames**. Show all your workings.

i. FIFO Page Replacement [5 marks]

ii. OPT (Optimal) Page Replacement [5 marks]

iii. LRU (Least Recently Used) Page Replacement [5 marks]

(b) Which page replacement algorithm demonstrates the best performance based on the number of page faults. [2 mark]

QUESTION 5 [15 MARKS]

(a) **Table 4** shows a bit map for free space management for an operating system where a bit 0 indicates that the block is free and bit 1 indicates that the block is occupied. Let's assume that each block size is 10KB.

Table 4

Block Number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bit Map	0	1	0	0	1	0	1	1	0	1	0	1	0	0	1	0

i) If a file **RECORDS.bin** is **35 KB** in size, determine which block numbers will be utilized to store the file. Provide your calculations. [2 marks]

ii) Upon examining the allocation for **RECORDS.bin** as identified in part (i), is there any internal fragmentation? If present, calculate the fragmentation size. If not, briefly justify your reasoning. [2 marks]

(b) The system has a disk with 16 storage blocks, each block's size being 8KB, as illustrated in **Figure 6**.

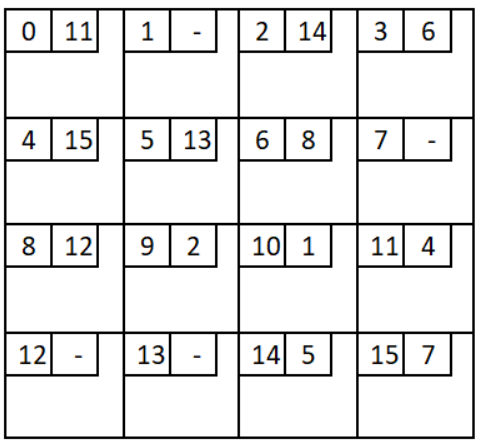

Figure 6

Files **Report.doc** and **Presentation.ppt** use a linked allocation method starting at blocks **3** and **9**, respectively. The blocks used by the files are scattered throughout the disk. Address the following inquiries:

i) List down by sequence all blocks used by each file. [2 marks]
ii) Calculate the total size occupied by each file, assuming no fragmentation has occurred. Show your calculations. [2 marks]

(c) Consider a file system that uses an indexing scheme to manage its free space, with the following list of free block addresses:

5, 10, 15, 20, 25, 2, 7, 12, 17, 22, 3, 8, 13, 18, 23, 4, 9, 14, 19, 24

Given that each index block can store up to four block addresses and employs a linked scheme:

i) Identify the index blocks and list their contents. [2 marks]
ii) Sketch a diagram showing the index blocks and their respective linked blocks. [5 marks]